Continuum Mechanics

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Fall 2026

Chapter 01

Tensor Algebra

Part 1 of 5

Course Reference

We will follow the textbook

A First Course in Continuum Mechanics
O. Gonzalez and A. M. Stuart, Cambridge University Press, 2008.

Please read Section 1.1 independently. We begin with Section 1.2.

Points and Vectors

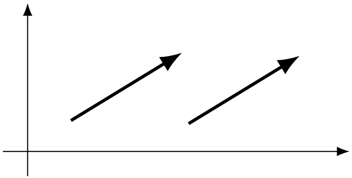
The book distinguishes between

$$\mathbb{E}^3 = \{\text{points in Euclidean three-space}\}$$

and

$$\mathcal{V} = \{\text{vectors in Euclidean three-space}\}.$$

Points are fixed locations. Vectors have magnitude and direction and may be translated without changing the vector.



These two arrows represent the same vector.

Section 1.2
Index Notation

The Canonical Basis

The canonical basis vectors in $\mathcal{V}$ are

$$\hat{\mathbf{e}}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \hat{\mathbf{e}}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \hat{\mathbf{e}}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

For

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix},$$

the jth component is written

$$v_j = [\mathbf{v}]_j, \qquad j = 1, 2, 3.$$

Similarly,

$$[\hat{\mathbf{e}}_i]_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

The Kronecker Delta

Definition

The *Kronecker delta* is

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \qquad i, j \in \{1, 2, 3\}.$$

Thus,

$$[\widehat{\mathbf{e}}_i]_j = \delta_{ij}.$$

Check this directly for each pair $i, j \in \{1, 2, 3\}$.

The Summation Convention

Every vector $\mathbf{v} \in \mathcal{V}$ can be written as

$$\mathbf{v} = \sum_{i=1}^{3} v_i \widehat{\mathbf{e}}_i .$$

Using the summation convention, we write this more compactly as

$$\mathbf{v} = v_i \widehat{\mathbf{e}}_i .$$

Definition

Under the *summation convention*, summation from 1 to 3 is implied whenever an index is repeated in a term.

The repeated index is a *dummy index*; it may be renamed without changing the expression.

Free and Dummy Indices

In the expression

$$\mathbf{v} = v_i \widehat{\mathbf{e}}_i ,$$

i is a *dummy index*: it is repeated, so summation is implied. A dummy index may be renamed without changing the expression:

$$v_i \widehat{\mathbf{e}}_i = v_j \widehat{\mathbf{e}}_j .$$

In the component identity

$$v_j = [\mathbf{v}]_j ,$$

j is a *free index*: it is not summed. Free indices must agree in every term of an equation. Within a single term, an index should not appear more than twice.

Example

The summation convention gives

$$\mathbf{v} = v_i \widehat{\mathbf{e}}_i = v_1 \widehat{\mathbf{e}}_1 + v_2 \widehat{\mathbf{e}}_2 + v_3 \widehat{\mathbf{e}}_3.$$

For a specific free index, the Kronecker delta selects one component:

$$\begin{aligned} \delta_{2j} v_j &= \delta_{21} v_1 + \delta_{22} v_2 + \delta_{23} v_3 \\ &= 0 v_1 + 1 v_2 + 0 v_3 \\ &= v_2. \end{aligned}$$

Here 2 is free and j is dummy.

Orthonormal Frames

More generally, let

$$\mathcal{F} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

be an orthonormal frame:

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}.$$

If, in addition,

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \qquad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \qquad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2,$$

then $\mathcal{F}$ is a *right-handed orthonormal frame*.

For every $\mathbf{v} \in \mathcal{V}$, there are unique scalars v_1, v_2, v_3 such that

$$\mathbf{v} = v_i \mathbf{e}_i.$$

The components depend on the chosen frame; the vector does not.

The Vector Operations Used in This Lecture

We do not yet need geometric definitions of the dot and cross products. For the calculations below, we use their linearity in each argument. For example,

$$(\alpha \mathbf{a} + \beta \mathbf{b}) \cdot \mathbf{c} = \alpha(\mathbf{a} \cdot \mathbf{c}) + \beta(\mathbf{b} \cdot \mathbf{c}),$$
$$(\alpha \mathbf{a} + \beta \mathbf{b}) \times \mathbf{c} = \alpha(\mathbf{a} \times \mathbf{c}) + \beta(\mathbf{b} \times \mathbf{c}),$$

with analogous identities in the second argument.

We also need their action on frame vectors. For an orthonormal frame,

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}.$$

For a right-handed orthonormal frame,

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \qquad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \qquad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2,$$
$$\mathbf{e}_2 \times \mathbf{e}_1 = -\mathbf{e}_3, \qquad \mathbf{e}_3 \times \mathbf{e}_2 = -\mathbf{e}_1, \qquad \mathbf{e}_1 \times \mathbf{e}_3 = -\mathbf{e}_2,$$

and $\mathbf{e}_i \times \mathbf{e}_i = \mathbf{0}$. The permutation symbol introduced later will encode these nine rules compactly.

Dot Products in Component Form

Let $\mathbf{a} = a_i \mathbf{e}_i$ and $\mathbf{b} = b_i \mathbf{e}_i$, where $\mathcal{F}$ is orthonormal. To keep the two sums distinct, first write

$$\mathbf{a} = \sum_{i=1}^{3} a_i \mathbf{e}_i, \qquad \mathbf{b} = \sum_{j=1}^{3} b_j \mathbf{e}_j.$$

Then

$$\begin{aligned}
\mathbf{a} \cdot \mathbf{b} &= \left(\sum_{i=1}^{3} a_i \mathbf{e}_i \right) \cdot \left(\sum_{j=1}^{3} b_j \mathbf{e}_j \right) \\
&= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j (\mathbf{e}_i \cdot \mathbf{e}_j) \\
&= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j \delta_{ij}.
\end{aligned}$$

Dot Products in Component Form

Continuing the calculation,

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j \delta_{ij}$$
$$= \sum_{i=1}^{3} a_i b_i$$
$$= a_i b_i,$$

where the last line uses the summation convention.

Only the terms with $i = j$ survive:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

The Permutation Symbol

Definition

For a right-handed orthonormal frame, define

$$\epsilon_{ijk} = (\mathbf{e}_i \times \mathbf{e}_j) \cdot \mathbf{e}_k .$$

Equivalently,

$$\epsilon_{ijk} = \begin{cases} +1, & (i,j,k) = (1,2,3),(2,3,1),(3,1,2), \\ -1, & (i,j,k) = (3,2,1),(2,1,3),(1,3,2), \\ 0, & \text{otherwise.} \end{cases}$$

The symbols δ_{ij} and ϵ_{ijk} are unchanged by a change from one right-handed orthonormal frame to another.

Alternating and Cyclic Properties

Proposition

For any $i, j, k \in \{1, 2, 3\}$, *interchanging any two indices reverses the sign:*

$$\epsilon_{ijk} = -\epsilon_{jik}, \qquad \epsilon_{ijk} = -\epsilon_{ikj}, \qquad \epsilon_{ijk} = -\epsilon_{kji}.$$

Cyclic permutations preserve the sign:

$$\epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij}.$$

Proposition

The permutation symbol is the determinant

$$\epsilon_{ijk} = \det\left(\begin{bmatrix} | & | & | \\ \mathbf{e}_i & \mathbf{e}_j & \mathbf{e}_k \\ | & | & | \end{bmatrix}\right).$$

Two Fundamental Frame Identities

Proposition

For every orthonormal frame $\{\mathbf{e}_i\}$,

$$\mathbf{e}_i = \delta_{ij}\mathbf{e}_j .$$

For every right-handed orthonormal frame,

$$\mathbf{e}_i \times \mathbf{e}_j = \epsilon_{ijk}\mathbf{e}_k .$$

Both identities use the summation convention.

Proof.

We want to show that

$$\mathbf{e}_i = \delta_{ij}\mathbf{e}_j.$$

Expanding the implied sum gives

$$\delta_{ij}\mathbf{e}_j = \delta_{i1}\mathbf{e}_1 + \delta_{i2}\mathbf{e}_2 + \delta_{i3}\mathbf{e}_3.$$

We now check the three possible values of the free index i.

Proof (Cont.)

If $i = 1$, then

$$\begin{aligned}\delta_{1j}\mathbf{e}_j &= \delta_{11}\mathbf{e}_1 + \delta_{12}\mathbf{e}_2 + \delta_{13}\mathbf{e}_3 \\ &= 1\mathbf{e}_1 + 0\mathbf{e}_2 + 0\mathbf{e}_3 \\ &= \mathbf{e}_1 .\end{aligned}$$

If $i = 2$, then

$$\begin{aligned}\delta_{2j}\mathbf{e}_j &= \delta_{21}\mathbf{e}_1 + \delta_{22}\mathbf{e}_2 + \delta_{23}\mathbf{e}_3 \\ &= 0\mathbf{e}_1 + 1\mathbf{e}_2 + 0\mathbf{e}_3 \\ &= \mathbf{e}_2 .\end{aligned}$$

Proof (Cont.)

Finally, if $i = 3$, then

$$\begin{aligned}\delta_{3j}\mathbf{e}_j &= \delta_{31}\mathbf{e}_1 + \delta_{32}\mathbf{e}_2 + \delta_{33}\mathbf{e}_3\\ &= 0\mathbf{e}_1 + 0\mathbf{e}_2 + 1\mathbf{e}_3\\ &= \mathbf{e}_3.\end{aligned}$$

Therefore,

$$\mathbf{e}_i = \delta_{ij}\mathbf{e}_j, \qquad i = 1, 2, 3.$$

□

The Transfer Property

The identity

$$\delta_{ij}\mathbf{e}_j = \mathbf{e}_i$$

is one instance of the transfer property:

$$\delta_{ij} v_j = v_i .$$

Exercise

Show that, for any right-handed orthonormal frame,

$$\mathbf{e}_i = \frac{1}{2}\epsilon_{ijk}(\mathbf{e}_j \times \mathbf{e}_k).$$

Vector Operations in Component Form

Proposition

Let

$$\mathbf{a} = a_i \mathbf{e}_i, \qquad \mathbf{b} = b_i \mathbf{e}_i, \qquad \mathbf{c} = c_i \mathbf{e}_i.$$

Then

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i,$$

$$\mathbf{a} \times \mathbf{b} = \epsilon_{ijk} a_i b_j \mathbf{e}_k,$$

and

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \epsilon_{ijk} a_i b_j c_k. \tag{1.2}$$

The three repeated indices in the final expression indicate three sums.

Proof.

First,

$$\begin{aligned}\mathbf{a} \cdot \mathbf{b} &= (a_i \mathbf{e}_i) \cdot (b_j \mathbf{e}_j) \\ &= a_i b_j (\mathbf{e}_i \cdot \mathbf{e}_j) \\ &= a_i b_j \delta_{ij} \\ &= a_i b_i .\end{aligned}$$

The transfer property of δ_{ij} replaces the index j by i.

Proof (Cont.)

Next,

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= (a_i \mathbf{e}_i) \times (b_j \mathbf{e}_j) && \text{(two sums)} \\ &= a_i b_j (\mathbf{e}_i \times \mathbf{e}_j) && \text{(two sums)} \\ &= a_i b_j \epsilon_{ijk} \mathbf{e}_k && \text{(three sums).} \end{aligned}$$

To see what the summation convention has saved us, we redo this calculation with every sum displayed.

Proof (Cont.)

$$
\begin{aligned}
\mathbf{a} \times \mathbf{b} &= \left(\sum_{i=1}^{3} a_i \mathbf{e}_i \right) \times \left(\sum_{j=1}^{3} b_j \mathbf{e}_j \right) \\
&= \sum_{i=1}^{3} a_i \mathbf{e}_i \times \left(\sum_{j=1}^{3} b_j \mathbf{e}_j \right) \\
&= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j (\mathbf{e}_i \times \mathbf{e}_j) \\
&= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j \left(\sum_{k=1}^{3} \epsilon_{ijk} \mathbf{e}_k \right) \\
&= \sum_{i=1}^{3} \sum_{j=1}^{3} \sum_{k=1}^{3} a_i b_j \epsilon_{ijk} \mathbf{e}_k .
\end{aligned}
$$

Proof (Cont.)

Set

$$\mathbf{d} = \mathbf{a} \times \mathbf{b}.$$

Since

$$\mathbf{d} = d_k \mathbf{e}_k \qquad \text{and} \qquad \mathbf{a} \times \mathbf{b} = a_i b_j \epsilon_{ijk} \mathbf{e}_k ,$$

uniqueness of components gives

$$d_k = [\mathbf{d}]_k = a_i b_j \epsilon_{ijk} , \qquad k = 1, 2, 3.$$

Now apply the dot-product rule:

$$\begin{aligned}(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} &= d_k c_k = [\mathbf{d}]_k c_k \\ &= a_i b_j c_k \epsilon_{ijk} ,\end{aligned}$$

which is equation (1.2). $\square$

Example

For a right-handed orthonormal frame,

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3.$$

Antisymmetry gives

$$\mathbf{e}_2 \times \mathbf{e}_1 = -\mathbf{e}_3,$$

while

$$\mathbf{e}_i \times \mathbf{e}_i = \mathbf{0}, \qquad i = 1, 2, 3.$$

Thus, reversing the order reverses the direction, and crossing a vector with itself gives the zero vector.

Example

Let

$$\mathbf{a} = \mathbf{e}_1 + 2\mathbf{e}_2, \qquad \mathbf{b} = 3\mathbf{e}_2 + \mathbf{e}_3.$$

The determinant mnemonic used in calculus gives

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ 1 & 2 & 0 \\ 0 & 3 & 1 \end{vmatrix}.$$

Expanding along the first row,

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \mathbf{e}_1 \begin{vmatrix} 2 & 0 \\ 3 & 1 \end{vmatrix} - \mathbf{e}_2 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \mathbf{e}_3 \begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix} \\ &= 2\mathbf{e}_1 - \mathbf{e}_2 + 3\mathbf{e}_3. \end{aligned}$$

Example (Cont.)

We now compute the same result using the permutation symbol:

$$[\mathbf{a} \times \mathbf{b}]_k = \epsilon_{ijk} a_i b_j .$$

The three components are

$$\begin{aligned}
[\mathbf{a} \times \mathbf{b}]_1 &= \epsilon_{231} a_2 b_3 + \epsilon_{321} a_3 b_2 = 2 + 0 = 2, \\
[\mathbf{a} \times \mathbf{b}]_2 &= \epsilon_{132} a_1 b_3 + \epsilon_{312} a_3 b_1 = -1 + 0 = -1, \\
[\mathbf{a} \times \mathbf{b}]_3 &= \epsilon_{123} a_1 b_2 + \epsilon_{213} a_2 b_1 = 3 + 0 = 3.
\end{aligned}$$

Therefore,

$$\mathbf{a} \times \mathbf{b} = 2\mathbf{e}_1 - \mathbf{e}_2 + 3\mathbf{e}_3 .$$

Proposition

For arbitrary $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathcal{V}$,

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} = (\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b}.$$

Moreover,

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \det\left(\begin{bmatrix} | & | & | \\ \mathbf{a} & \mathbf{b} & \mathbf{c} \\ | & | & | \end{bmatrix}\right). \tag{1.3}$$

Proof.

Exercise. $\square$

Combining Equations (1.2) and (1.3)

Equation (1.2) gives

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \epsilon_{ijk} a_i b_j c_k .$$

Equation (1.3) gives

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \det\left(\begin{bmatrix} | & | & | \\ \mathbf{a} & \mathbf{b} & \mathbf{c} \\ | & | & | \end{bmatrix} \right) .$$

Consequently,

$$\det\left(\begin{bmatrix} | & | & | \\ \mathbf{a} & \mathbf{b} & \mathbf{c} \\ | & | & | \end{bmatrix} \right) = \epsilon_{ijk} a_i b_j c_k . \tag{1.4}$$

The Vector Triple Product in Components

Proposition (Triple Vector Product)

Let

$$\mathbf{a} = a_q \mathbf{e}_q, \qquad \mathbf{b} = b_i \mathbf{e}_i, \qquad \mathbf{c} = c_j \mathbf{e}_j.$$

Then

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \epsilon_{pqk} \epsilon_{ijk} a_q b_i c_j \mathbf{e}_p. \tag{1.5}$$

Proof.

Let

$$\mathbf{d} = \mathbf{b} \times \mathbf{c}.$$

From the component formula for the cross product,

$$d_k = [\mathbf{d}]_k = \epsilon_{ijk} b_i c_j,$$

and hence

$$\mathbf{d} = \epsilon_{ijk} b_i c_j \mathbf{e}_k.$$

Therefore,

$$\begin{aligned}
\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (a_q \mathbf{e}_q) \times (\epsilon_{ijk} b_i c_j \mathbf{e}_k) \\
&= \epsilon_{ijk} a_q b_i c_j (\mathbf{e}_q \times \mathbf{e}_k) \\
&= \epsilon_{ijk} a_q b_i c_j \epsilon_{qkp} \mathbf{e}_p \\
&= \epsilon_{pqk} \epsilon_{ijk} a_q b_i c_j \mathbf{e}_p.
\end{aligned}$$

□

Why Products of Permutation Symbols Matter

Expressions containing two cross products naturally produce a product of two permutation symbols. Result 1.1 replaces that product by Kronecker deltas:

$$\epsilon_{pqk}\epsilon_{ijk} = \delta_{pi}\delta_{qj} - \delta_{pj}\delta_{qi}.$$

The transfer property then contracts the Kronecker deltas. Thus, a complicated vector expression can be reduced to dot products and scalar components by straightforward algebra.

Result 1.1 (Epsilon-Delta Identities)

$$\epsilon_{pqk}\epsilon_{ijk} = \delta_{pi}\delta_{qj} - \delta_{pj}\delta_{qi}.$$

and

$$\epsilon_{pqk}\epsilon_{iqk} = 2\delta_{pi}.$$

Proof.

Exercise. □

Applying Result 1.1 to Equation (1.5)

From equation (1.5),

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \epsilon_{pqk}\epsilon_{ijk} a_q b_i c_j \mathbf{e}_p.$$

Result 1.1 gives

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (\delta_{pi}\delta_{qj} - \delta_{pj}\delta_{qi}) a_q b_i c_j \mathbf{e}_p \\ &= \delta_{pi}\delta_{qj} a_q b_i c_j \mathbf{e}_p - \delta_{pj}\delta_{qi} a_q b_i c_j \mathbf{e}_p. \end{aligned}$$

Applying Result 1.1 to Equation (1.5)

We apply the transfer property to each term:

$$\begin{aligned}\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= a_q b_p c_q \mathbf{e}_p - a_q b_q c_p \mathbf{e}_p \\ &= (a_q c_q) b_p \mathbf{e}_p - (a_q b_q) c_p \mathbf{e}_p \\ &= (\mathbf{a} \cdot \mathbf{c})\, b_p \mathbf{e}_p - (\mathbf{a} \cdot \mathbf{b})\, c_p \mathbf{e}_p \\ &= (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.\end{aligned}$$

Therefore,

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

Principal Component Identities

For a right-handed orthonormal frame,

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i,$$

$$\mathbf{a} \times \mathbf{b} = \epsilon_{ijk} a_i b_j \mathbf{e}_k,$$

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \epsilon_{ijk} a_i b_j c_k,$$

and

$$\epsilon_{pqk}\epsilon_{ijk} = \delta_{pi}\delta_{qj} - \delta_{pj}\delta_{qi}.$$

These identities reduce vector operations to systematic scalar component calculations.